Transilvania University of Brasov

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Verification Project Environmental Control Device

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1 THE DIGITAL DESIGN OF THE CIRCUIT

1.1 BLOCK DIAGRAM

Figure 1. Block diagram of the device

1.2 DESCRIPTION OF THE GENERAL FUNCTIONALITY OF THE DUT (DEVICE UNDER TEST)

The environmental controller receives temperature, humidity and light intensity measurements from sensors and controls heating, air conditioning, a dehumidifier and automatic blinds.

The button behaves as a toggle switch. When pressed, Enable becomes 1 and the controller starts operating. When pressed again, Enable becomes 0 and the controller is disabled.

Table 1. Sensor ranges

Sensor measurement	Min Value	Max Value
Temperature	0	40
Humidity	0	100
Light Intensity	0	900

Based on the temperature sensor, the heating system turns on when the temperature drops below 22 degrees Celsius. The air conditioner turns on when the temperature exceeds 25 degrees Celsius and turns off when it reaches 22 degrees.

The values received from the humidity sensor control the dehumidifier. It turns on at a value exceeding 50% and turns off at values between 0% and 35%.

Depending on the light intensity levels, the curtains are opened or closed to limit the amount of light. The recommended light level in an office is 300–500 lux; thus, when the threshold of 700 lux is exceeded, the curtain closing system will activate, and at 200 lux, the curtains will open.

If the button is not pressed (enable=0) or the reset is active (reset_n = 1), all DUT outputs are set to 0. After the button is pressed again (enable = 1), the DUT outputs will be updated only after new data is read from the sensors.

Data received when the module's enable signal is 0 is ignored (OFF state). Data is read only when enable, valid and ready signals are all asserted (load state). In the control state, various commands are executed based on the information received from the sensors.

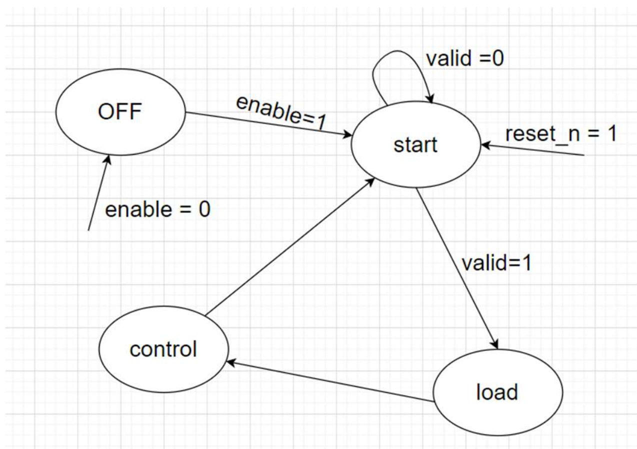

Figure 2. Finite State Machine describing the functionality of the device

1.3 DUT INTERFACES

Table 2. System signals generated in the testbench

General Signals			
Signal Name	I/O	Size	Description
clk_i	I	1	Clock signal
reset_i	I	1	Asynchronous reset, active high

Table 3. Button Interface

Button Interface			
Signal Name	I/O	Size	Description
enable_i	I	1	For enabling/disabling device

Table 4. Sensor Interface

Sensor Interface			
Signal Name	I/O	Size	Description
temperature_i	I	6	Temperature data
humidity_i	I	7	Humidity data
luminous_intensity_i	I	10	Luminous intensity data
valid_i	I	1	Valid signal
ready_o	O	1	Ready signal

Table 5. Actuator Interface

Actuator Interface			
Signal Name	I/O	Size	Description
heat_o	O	1	Heating status
AC_o	O	1	AC status
dehumidifier_o	O	1	Dehumidifier status
blinds_o	O	1	Blinds status

1.4 List of DUT Functionalities

- Reads button state for enabling/disabling
- Receives environmental data from sensors
- Processes sensor data
- Enables/disables heat, AC, dehumidifier and blinds

1.5 Waveforms

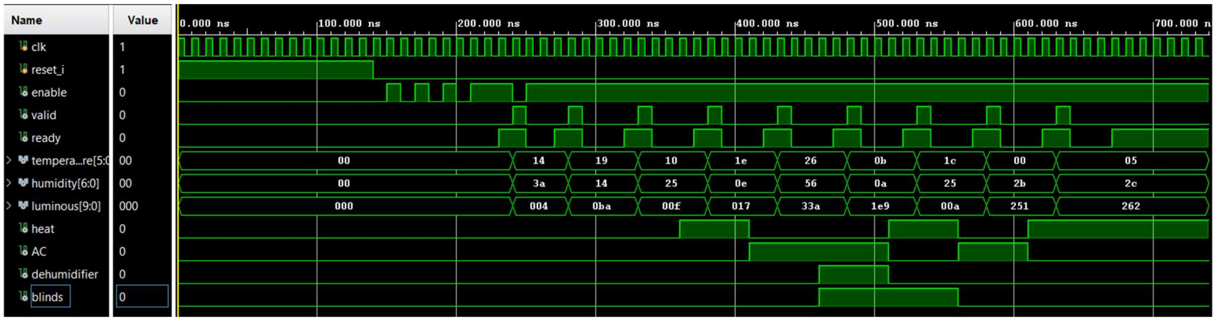

Figure 3. Waveforms from the DUT

2 DIGITAL CIRCUIT VERIFICATION

2.1 Verification Environment Architecture

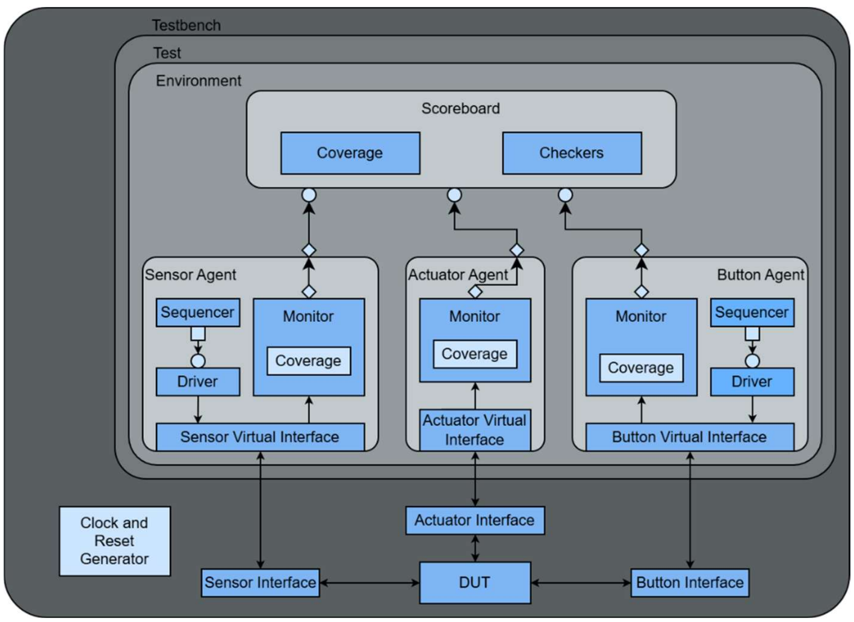

Figure 4. Overview of the components necessary for sending stimuli to the DUT

2.1.1 Sensor Agent

The Sensor Agent is an active agent that encapsulates the Sequencer, Driver and Monitor into a single entity, connecting the components with the DUT using the virtual interface.

2.1.2 Sensor Transaction

Table 6. Sensor Transaction

Field Name	Type	Size	Values	Description
temperature	rand bit	6	[0:40]	Value of environmental temperature
humidity	rand bit	7	[0:100]	Value of environmental humidity
luminous_intensity	rand bit	10	[0:900]	Value of environmental light intensity

2.1.3 Sensor Driver

The Sensor Driver interacts with the DUT via the Sensor Interface, sending the transactions generated by the Sensor Sequencer. The transactions follow the valid-ready protocol.

The driver receives transactions from the sequencer, waits for the ready signal to assert, asserts valid signal and drives sensors values, and after a clock cycle de-asserts the valid signal.

2.1.4 Sensor Monitor

The Sensor Monitor waits for valid and ready signals to be both asserted, samples the sensors values from the Sensor Virtual Interface, reconstructs the transaction and sends it to the Scoreboard and Sensor Coverage Collector.

2.1.5 Sensor Coverage Collector

The Sensor Coverage Collector is instantiated inside the Sensor Monitor and receives transactions from it. The coverage collector contains sensor_cg cover group with 4 cover points: temperature_cp, ac_cp, humidity_cp, luminous_intensity_cp.

2.1.6 Sensor Sequences

The Sensor Sequences generate stimulus for the Sensor Driver. Each sequence creates one or more constrained-random transactions that the driver sends to Sensor Interface. The following sequences are implemented:

Table 7. Sensor Sequences

Sequence	Description
temperature_sequence	Sends values of environmental temperature for every state of heat and AC.
humidity_sequence	Sends values of environmental humidity for every state of dehumidifier.
luminosity_sequence	Sends values of environmental light intensity for every state of blinds.
sensor_limit_values_sequence	Sends only lower and upper limits of every environmental data range.
random_sensor_sequence	Sends random values of environmental data.

2.1.7 Actuator Agent

The Actuator Agent is a passive agent that contains only the Monitor and connects it with the DUT using the virtual interface. The Actuator Interface has only output signals from DUT, meaning no driver or sequencer is needed.

2.1.8 Actuator Transaction

Table 8. Actuator Transaction

Field Name	Type	Size	Values	Description
Heat_i	rand bit	1	[0:1]	Heat command
AC_i	rand bit	1	[0:1]	AC command
Blinds_i	rand bit	1	[0:1]	Blinds command
Dehumidifier_i	rand bit	1	[0:1]	Dehumidifier command

2.1.9 Actuator Monitor

The Actuator Monitor samples the commands sent by DUT every time each of the values changes, reconstructs the transaction and sends it to the Scoreboard and Actuator Coverage Collector.

2.1.10 Actuator Coverage Collector

The Actuator Coverage Collector is instantiated inside the Actuator Monitor and receives transactions from it. The coverage collector contains actuator_cg cover group with 4 coverpoints: heat, ac, blinds, dehumidifier; and actuator_cross cross.

2.1.11 Button Agent

The Button Agent is an active agent that encapsulates the Sequencer, Driver and Monitor into a single entity, connecting the components with the DUT using the virtual interface.

2.1.12 Button Transaction

Table 9. Button Transaction

Field Name	Type	Size	Values	Description
enable	rand bit	1	[0:1]	Value of enable signal

2.1.13 Button Driver

The Button Driver receives transactions from the Button Sequencer and drives the enable signal.

2.1.14 Button Monitor

The Button Monitor waits for the enable signal to change, reconstructs the transaction and sends it to the Scoreboard and Button Coverage Collector.

2.1.15 Button Coverage Collector

The Button Coverage Collector is instantiated inside the Button Monitor and receives transactions from it. The coverage collector contains button_cg cover group with button_state cover point.

2.1.16 Button Sequences

The Button Sequences generate stimulus for the Button Driver. Each sequence creates one or more constrained-random transactions that the driver sends to Button Interface. The following sequences are implemented:

Table 10. Button Sequences

Sequence	Description
fast_switch_sequence	Alternative values for enable signal.
frequent_button_sequence	Device is mostly enabled.
rare_button_sequence	Device is mostly disabled.
button_sequence	Random values for enable signal.

2.1.17 Scoreboard

The Scoreboard receives transactions from all monitors.

When receiving a Button Transaction, the scoreboard stores the value of enable signal and predicts an implicit transaction from Sensor Monitor if enable is 0.

When receiving a Sensor Transaction, predicts the next Actuator Transaction if the stored enable signal value is 1.

All predicted Actuator Transactions are stored in a queue.

When receiving an Actuator Transaction, the scoreboard compares it with the predicted transaction, generating an error message if the checker fails.

2.1.18 Scoreboard Coverage Collector

The Scoreboard Coverage Collector is instantiated inside the Scoreboard and receives transactions from it. The coverage collector contains processed_data_cg cover group with 4 cover points: scbd_enable_cp, scbd_temperature_cp, scbd_humidity_cp, scbd_luminous_intensity_cp; and 3 crosses: temperature_cross, humidity_cross, luminous_intensity_cross.

2.2 VERIFICATION CHECKLIST

2.2.1 Assertions

All Sensor assertions are disabled when reset is active to prevent false assertion failures.

Table 11. Actuator Interface Assertions

Assertion	Description
temperature_control	Verifies that heat and AC are not on at the same time.

Table 12. Sensor Interface Assertions

Assertion	Description
temperature_range	Verifies that temperature signal is in the interval [0,40].
humidity_range	Verifies that humidity signal is in the interval [0,100].
luminous_intensity_range	Verifies that luminous intensity signal is in the interval [0,900].

2.3 FUNCTIONAL COVERAGE ELEMENTS

Table 13. Actuator cover group

actuator_cg		
Cover point	Values	Description
heat	[0:1]	Covers heat command
ac	[0:1]	Covers AC command
blinds	[0:1]	Covers blinds command
dehumidifier	[0:1]	Covers dehumidifier command
actuator_cross	heat x ac x blinds x dehumidifier	Covers all combinations of commands

Table 14. Sensor cover group

button_cg		
Cover point	Values	Description
button_state	[0:1]	Covers device state

Table 15. Sensor cover group

sensor_cg			
Cover point	Bins	Values	Description
temperature_cp	heat_on	[0:21]	Temperature for heat on
	heat_limit_on	22	Temperature limit value for heat on
	heat_limit_off	23	Temperature limit value for heat off
	heat_off	[24:40]	Temperature for heat off
ac_cp	ac_off	[0:21]	Temperature for ac off
	ac_limit_off	22	Temperature limit value for ac off
	ac_stable	[23:24]	Temperature for ac stable
	ac_limit_on	25	Temperature limit value for ac on
	ac_on	[26:40]	Temperature for ac on
humidity_cp	dehumidifier_off	[0:34]	Humidity for dehumidifier off
	dehumidifier_limit_off	35	Humidity limit value for dehumidifier off
	dehumidifier_stable	[36:49]	Humidity value for dehumidifier stable
	dehumidifier_limit_on	50	Humidity limit value for dehumidifier on
	dehumidifier_on	[51:100]	Humidity for dehumidifier on

luminous _intensity_cp	blinds_open	[0:199]	Luminous intensity for blinds open
	blinds_limit_open	200	Luminous intensity limit value for blinds open
	blins_stable	[201:699]	Luminous intensity for blinds stable
	blinds_limit_close	700	Luminous intensity limit value for blinds closed
	blinds_closed	[701:900]	Luminous intensity for blinds closed

Table 16. Scoreboard cover group

processed_data_cg		
Cover point	Values	Description
scbd_enable_cp	[0:1]	Covers device state
scbd_temperature_cp	[0:40]	Covers temperature values
scbd_humidity_cp	[0:100]	Covers humidity values
scbd_luminous _intensity_cp	[0:900]	Covers luminous intensity values
temperature_cross	scbd_enable_cp x scbd_temperature_cp	Covers temperature values when device is enabled
humidity_cross	scbd_enable_cp x scbd_humidity_cp	Covers humidity values when device is enabled
luminous_intensity _cross	scbd_enable_cp x scbd_luminous_intensity_cp	Covers luminous intensity values when device is enabled

2.4 VERIFICATION SCENARIOS AND TESTS

Table 17. Verification Scenarios

Scenario	Description
Reset output values	Verifies that after a reset signal is applied, all DUT outputs are initialized to their expected default values.
Random Sensor Verification	Verifies the correct operation of the temperature, humidity, and light sensors using randomly generated input values.
Boundary Value Verification	Assign fixed boundary sensor values defined in the specification table.
Humidity Control Verification	Verifies that the humidity control logic operates correctly when humidity values remain within the normal operating range.
Light Intensity Control Verification	Verifies correct DUT behavior when light intensity values are within the recommended range.
Temperature Control Verification	Verifies the temperature control functionality when temperature values remain between 22 °C and 25 °C.
Enable Signal Functionality Verification	Verifies correct DUT operation when the enable signal is repeatedly toggled through multiple button presses.
Button Functionality Verification	Verifies the correct response of the DUT to rapid button presses and ensures proper button handling.
Periodic Shutdown Verification	Verifies that the environmental control module can be disabled and re-enabled during operation.

Table 18. List of implemented tests for DUT verification

Test file name	Test description
frequent_button_test	Test that DUT enables and disables correctly when button is pressed rapidly.
humidity_test	Test the DUT behavior for different humidity values from all operating ranges
luminosity_test	Test the DUT behavior for different luminosity values from all operating ranges

temperature_test	Test the DUT behavior for different temperature values from all operating ranges
sensor_limit_values_test	Test that DUT behaves correctly when sensor inputs are set to boundary values.
sensor_test	Test the correct operation of the environmental control system using randomly generated temperature, humidity, and light intensity values.

2.5 Results

Running the regression produced the following results:

- Sensor functional coverage: 100%
- Actuator functional coverage: 100%
- Button functional coverage: 100%
- Scoreboard coverage: 100%

All assertions passed without failures throughout the simulation